

Simulation design: asymmetric-KITE gridium and the 3-photon, 2-frequency R_x gate

Design document — v1

1 Objective and scope

The logical R_x on gridium is synthesized by a 3-photon, 2-frequency flux drive through the bright higher levels, ladder $|0\rangle \rightarrow |5\rangle \rightarrow |4\rangle \rightarrow |1\rangle$. In the ideal symmetric device the higher levels pair into parity doublets that are degenerate at the protection bias, so the middle rung $|4\rangle \leftrightarrow |5\rangle$ is (i) at a near-DC frequency with no spectral selectivity and (ii) $\dim(|\langle 4|\hat{\varphi}|5\rangle| \sim 0.03$ in the effective 1-mode model). In the fabricated device the KITE is not perfectly symmetric; the asymmetry reintroduces a residual single-Cooper-pair (first-harmonic) term that lifts the doublet degeneracies and brightens the intra-doublet rungs.

This document specifies a simulation campaign to (1) quantify how the doublet degeneracies are lifted as the KITE asymmetries are varied, using the most device-accurate Hamiltonian available from the paper’s appendix (the four-mode circuit model with *exact* asymmetry terms, populated by the parameters extracted from the measured protected device); (2) locate the asymmetry/bias sweet spot for the 3-photon, 2-frequency R_x ; and (3) simulate, calibrate, and score the gate in the time domain by direct lab-frame integration of the Schrödinger equation (no rotating frame, no RWA).

Two facts sharpen the search. First, because the intermediate levels of the ladder are *virtual*, the three-photon resonance condition is independent of them:

$$\omega_a - \omega_b - \omega_a = E_1 - E_0 \implies \omega_b = -\delta_{01}, \quad \delta_{01} \equiv E_1 - E_0, \quad (1)$$

(two ω_a photons cancel: absorbed on $0 \rightarrow 5$, emitted on the descent $4 \rightarrow 1$), and the rotation axis is set by tone B’s phase alone, $\Phi = -\phi_b$. Second, asymmetry therefore helps the gate *twice*: it brightens $\langle 4| \cdot |5\rangle$, and it makes δ_{01} finite, moving ω_b from near-DC to a genuine, spectrally selective RF frequency. The cost is partial loss of protection. Quantifying this trade-off is the core deliverable.

2 Model: exact four-mode Hamiltonian with KITE asymmetry

2.1 Modes

Following SI Note 4 of the paper (branch-flux quantization of Fig. S4a): $\hat{\varphi}_\Sigma = (\hat{\phi}_1 + \hat{\phi}_2)/2$ is the common phase across the two KITE junctions (carries the Cooper-quartet dynamics); $\hat{\varphi}_\Delta = (\hat{\phi}_1 - \hat{\phi}_2)/2$ is the internal circulating mode of the rhombus, confined at $\vartheta_{\text{ext}}/2$ (the interference knob); $\hat{\varphi}_S$ is the phase across the small QPS junction (phase-slip dynamics); and $\hat{\phi}$ is the node between superinductor and KITE, with parasitic capacitance C_p (energy ε_p). Conjugate charges $\hat{n}_\Sigma, \hat{n}_\Delta, \hat{n}_S, \hat{n}_\phi$, with $[\hat{\varphi}_j, \hat{n}_j] = i$.

2.2 Asymmetry enters exactly through two cross terms

Junction asymmetry $E_{J1,2} = E_J(1 \mp \epsilon_J)$ obeys the identity

$$-E_{J1} \cos \hat{\phi}_1 - E_{J2} \cos \hat{\phi}_2 = -2E_J \cos \hat{\varphi}_\Sigma \cos \hat{\varphi}_\Delta - 2\epsilon_J E_J \sin \hat{\varphi}_\Sigma \sin \hat{\varphi}_\Delta, \quad (2)$$

exact to all orders in ϵ_J (paper Eq. S31). Inductor asymmetry $E_{LK1,2} = E_{LK}(1 \mp \epsilon_{LK})$: loop closure fixes the branch fluxes $\theta_{1,2} = a \pm b$ with $a = \hat{\varphi}_\Sigma - \hat{\phi}$, $b = \hat{\varphi}_\Delta - \vartheta_{\text{ext}}/2$ independently of element values, so

$$\frac{1}{2}E_{LK1}\theta_1^2 + \frac{1}{2}E_{LK2}\theta_2^2 = E_{LK}(a^2 + b^2) - 2\epsilon_{LK}E_{LK}ab, \quad (3)$$

also exact. (The paper's Eq. S32 is the first-order image of (3) after Born–Oppenheimer elimination of $\hat{\phi}$; simulating the four-mode model avoids that truncation entirely, which matters here because the fitted $\epsilon_p/h = 1$ GHz is *not* large compared to the other scales, so the BO elimination is marginal for the real device.)

2.3 The Hamiltonian to simulate

$$\hat{H} = 2E_C \hat{n}_\Sigma^2 + 2E_C^\Delta \hat{n}_\Delta^2 + 4E_{CS} \hat{n}_S^2 + 4\epsilon_p \hat{n}_\phi^2 \quad (4)$$

$$- 2E_J \cos \hat{\varphi}_\Sigma \cos \hat{\varphi}_\Delta - 2\epsilon_J E_J \sin \hat{\varphi}_\Sigma \sin \hat{\varphi}_\Delta - E_{JS} \cos \hat{\varphi}_S \quad (5)$$

$$+ E_{LK} \left[(\hat{\varphi}_\Sigma - \hat{\phi})^2 + (\hat{\varphi}_\Delta - \frac{\vartheta_{\text{ext}}}{2})^2 \right] - 2\epsilon_{LK} E_{LK} (\hat{\varphi}_\Sigma - \hat{\phi})(\hat{\varphi}_\Delta - \frac{\vartheta_{\text{ext}}}{2}) \quad (6)$$

$$+ E_L (\hat{\varphi}_S - \hat{\phi} + \varphi_{\text{ext}})^2, \quad (7)$$

with $E_C = e^2/2C_J$, $E_C^\Delta = e^2/2(C_J + \frac{1}{2}C_K)$ (cross-KITE capacitance ϵ_K , paper Eq. S30), $E_{CS} = e^2/2C_{JS}$, $\epsilon_p = e^2/2C_p$. Offset charge: the junction arrangement isolates a superconducting island; n_g is attached via the compact-variable transformation of Eqs. S24–S26 ($\hat{\varphi}_1 = \hat{\varphi}_S$ compact, $\hat{\varphi}_2 = \hat{\varphi}_S - \hat{\varphi}_\Sigma$ the extended grid mode, $\hat{\varphi}_3 = \hat{\varphi}_\Delta$), i.e. the charge dispersion is evaluated by sweeping n_g in the transformed frame. Protection bias: $(\varphi_{\text{ext}}, \vartheta_{\text{ext}}) = (0, \pi)$.

2.4 Drive and readout operators

RF on the differential flux line modulates the external fluxes, so the physically correct drive operators are the flux derivatives of (4)–(7), evaluated at the bias point:

$$\hat{D}_\vartheta = \frac{\partial \hat{H}}{\partial \vartheta_{\text{ext}}} = -E_{LK}(\hat{\varphi}_\Delta - \frac{\vartheta_{\text{ext}}}{2}) + \epsilon_{LK} E_{LK}(\hat{\varphi}_\Sigma - \hat{\phi}), \quad (8)$$

$$\hat{D}_\varphi = \frac{\partial \hat{H}}{\partial \varphi_{\text{ext}}} = 2E_L (\hat{\varphi}_S - \hat{\phi} + \varphi_{\text{ext}}), \quad (9)$$

with the measured crosstalk matrix (paper Eq. S35) mapping line currents (I_Δ, I_Σ) to $(\varphi_{\text{ext}}, \vartheta_{\text{ext}})$ drives; a capacitive drive couples through $\hat{n}_1 = \hat{n}_S + \hat{n}_\Sigma$. All operators are projected onto the lowest- M eigenbasis for the time-domain stage. Note that $\hat{D}_\vartheta$ itself acquires an asymmetry-induced piece, so the drive matrix elements must be recomputed at every $(\epsilon_J, \epsilon_{LK})$ point, not scaled from the symmetric case.

2.5 Parameters

Baseline: the four-mode parameters extracted from the measured protected device (paper SI Note 11), which is the closest available description of the actual hardware. Asymmetries are swept over the range measured across the wafer by room-temperature resistance probing (SI Note 12).

Parameter	Value ($h\cdot\text{GHz}$)	Source	Role
E_J	7.21	SI Note 11 fit	KITE junctions (mean)
E_C	0.77	SI Note 11 fit	KITE junction charging
E_L	0.73	SI Note 11 fit	superinductor
E_{LK}	0.73	SI Note 11 fit	KITE inductors (mean)
E_{JS}	1.14	SI Note 11 fit	QPS junction
E_{CS}	8.13	SI Note 11 fit	QPS charging
ε_K	0.68	SI Note 11 fit	cross-KITE capacitance
ε_p	1.0	SI Note 11 fit	parasitic node capacitance
E_C^Δ	0.492	derived (Eq. S30)	Δ -mode charging
ϵ_J	0–0.15 sweep	SI Note 12 probing	junction asymmetry
ϵ_{LK}	0–0.15 sweep	SI Note 12 probing	inductor asymmetry
n_g	$[0, 1)$ sweep	—	island offset charge

Caveat: fitted parameters are model-bound. These values were extracted with the four-mode model and must be used with it; the effective 1-mode parameters of Table S1 are *not* interchangeable with them. Fit uncertainties propagate mainly into transition frequencies, which the gate workflow recalibrates anyway (Step 3); the simulation’s job is the structure (which rungs are bright, how split), not MHz-accurate frequencies.

3 Numerical implementation

Construction. Build (4)–(7) either (a) in `scqubits.Circuit` from the branch list with per-branch $E_{J1,2}$, $E_{LK1,2}$ (asymmetry is then native and the compact/extended variable identification and hierarchical diagonalization are automatic), or (b) by hand as a four-fold tensor product. Basis per mode: charge basis for the compact island variable (~ 25 – 41 charge states); harmonic-oscillator bases for the extended modes — the grid combination needs the largest cutoff ($\gtrsim 300$), the Δ and ϕ modes are stiff (~ 15 – 25 and ~ 10 – 15 levels). Use sparse Lanczos for the lowest ~ 12 eigenpairs.

Convergence protocol (mandatory). The observables of interest (doublet splittings) are 10 – 10^3 MHz; truncation bias can be exactly that size. At three representative points — symmetric, $\epsilon_J = 3\%$, $\epsilon_J = 15\%$ — double every per-mode cutoff and require all reported splittings and matrix elements stable to $< 1\%$.

Step 0: validation before any sweep. (i) Symmetric limit: doublets $(0, 1)$, $(2, 3)$, $(4, 5)$ degenerate to numerical precision at the protection bias; spectrum-vs-flux reproduces the paper’s fitted spectra (Fig. 3f / Extended Data Fig. 3b). (ii) Cross-check the four-mode result against the three-mode model (Eq. S23 + S31 + S32) at small ϵ : agreement at first order in ϵ_{LK} , exact in ϵ_J . (iii) Parity bookkeeping: at $\epsilon = 0$, drive operators (8)–(9) obey the checkerboard selection rules (dark $0 \leftrightarrow 1$). (iv) n_g periodicity of the spectrum.

4 Step 1: degeneracy lifting vs asymmetry

At the protection bias, sweep a grid $(\epsilon_J, \epsilon_{LK}) \in [0, 0.15]^2$ (log-spaced near zero; include the axes $\epsilon_{LK} = 0$ and $\epsilon_J = 0$ to separate mechanisms). At each point record:

1. Doublet splittings $\delta_{01}, \delta_{23}, \delta_{45}$ and the outer-rung frequencies $E_5 - E_0$, $E_4 - E_1$ (their difference $= \delta_{45} + \delta_{01}$ sets how far tone A can sit from simultaneous outer-rung resonance).

2. Drive matrix elements $|\langle i|\hat{D}_\vartheta|j\rangle|$, $|\langle i|\hat{D}_\varphi|j\rangle|$, $|\langle i|\hat{n}_1|j\rangle|$ for all $i, j \leq 7$; in particular the middle rung (4,5), the outer rungs (0,5), (1,4), and the protection-critical (0,1).
3. Protection metrics: $|\langle 0|\hat{D}|1\rangle|$ for each drive channel (bit-flip/driving leakage), charge dispersion of δ_{01} and of the rung frequencies vs n_g , and local flux curvature $\partial^2\delta_{01}/\partial\vartheta_{\text{ext}}^2$, $\partial^2\delta_{01}/\partial\varphi_{\text{ext}}^2$ (dephasing).
4. Bias re-centering: under disorder the degeneracy/symmetry point shifts along the flux axes (observed in the paper, Extended Data Fig. 1); at each ϵ re-optimize $(\varphi_{\text{ext}}, \vartheta_{\text{ext}})$ locally before recording, and log the shift.

Expected physics to verify: the effective first-harmonic term is parity-*odd* at frustration ($\langle \sin \hat{\varphi}_\Delta \rangle \approx 1$ at $\hat{\varphi}_\Delta \approx \pi/2$, giving $\sim E_{1J} \sin \hat{\varphi}$ on the grid mode), so splittings grow linearly in ϵ_J ; the effective single-mode caricature gives $\delta_{45} \sim 10^2$ MHz already at $\epsilon_J \sim 1\%$ and a $\sim 10\times$ brightening of $|\langle 4|\cdot|5\rangle|$ by 3%, while the full multimode structure (island compactness, Δ -mode fluctuations) is expected to suppress these numbers substantially — the paper reports no appreciable spectral change up to 10% in its multimode simulations. Pinning the true suppression factor is the point of this step. Deliverable: maps of items 1–3 over the $(\epsilon_J, \epsilon_{LK})$ plane, plus the 1-mode-vs-4-mode comparison.

5 Step 2: sweet spot for the 3-photon, 2-frequency gate

Define the gate-level figures of merit from the Step-1 spectra, using resonance (1):

$$\begin{aligned} \Omega_X(\epsilon; \Delta_1) &\propto \frac{(\Omega_a d_{05})(\Omega_b d_{54})(\Omega_a d_{41})}{\Delta_1 \Delta_2}, & d_{ij} &\equiv |\langle i|\hat{D}|j\rangle|, & (10) \\ \mathcal{S}(\epsilon) &= \min(|\delta_{01}|, \text{gap to nearest spectator transition at } \omega_b) & & (\text{tone-B selectivity}), \\ \mathcal{P}(\epsilon) &= \left\{ |\langle 0|\hat{D}|1\rangle|, \max_{n_g} |\partial\delta_{01}|, \text{flux curvatures} \right\} & & (\text{protection costs}). \end{aligned}$$

Here Δ_1 is the tone-A detuning from the $0 \rightarrow 5$ rung and Δ_2 the accumulated two-photon detuning at $|4\rangle$; because $E_5 - E_0 \neq E_4 - E_1$ under asymmetry, tone A cannot sit symmetrically between the outer rungs — choose ω_a to balance the two virtual denominators (or optimize it numerically). The sweet spot is the constrained optimum: maximize predicted gate speed Ω_X subject to (i) $\mathcal{S}$ large enough that a pulse of duration $T \sim \pi/\Omega_X$ resolves ω_b from spectators ($\mathcal{S} \gg 1/T$), and (ii) every entry of $\mathcal{P}$ below an idle-error budget set by the target idle T_1, T_φ (thresholds to be agreed with the experiment; a reasonable opening stance is that asymmetry-induced idle error should not exceed the measured symmetric device's error). Output: the recommended $(\epsilon_J^*, \epsilon_{LK}^*)$ *acceptance window* — since fabrication sets ϵ , the practical deliverable is which measured wafer asymmetries (SI Note 12: 1–15%) fall inside the window, i.e. a device-selection rule, plus the re-centered bias point and the tone placement (ω_a, ω_b) for each acceptable device.

6 Step 3: time-domain gate simulation and calibration

Project $\hat{H}$ and the drive operators onto the lowest $M \sim 10$ eigenstates at the selected $(\epsilon^*, \text{bias})$. Total lab-frame Hamiltonian, no rotating frame, no RWA:

$$\hat{H}(t) = \hat{H}_0 + \hat{D} \left[\Omega_a \xi(t) \cos(\omega_a t + \phi_a) + \Omega_b \xi(t) \cos(\omega_b t + \phi_b) \right], \quad (11)$$

with smooth envelope $\xi(t)$ (raised-cosine or Gaussian flat-top). Integrate $i\partial_t U = \hat{H}(t)U$ with `qutip.propagator`; step-size controls must resolve the fast carrier ($\Delta t \ll 1/\omega_a$). Calibration loop, in order:

1. **Tone-B scan:** sweep ω_b around $-\delta_{01}$ (Eq. 1); the AC-Stark shift of the strong tone A displaces the true resonance — maximize end-of-pulse $P(0 \rightarrow 1)$.
2. **Amplitude/duration:** co-tune (Ω_a, Ω_b, T) for a π rotation; keep the intermediate-population transient bounded (leakage proxy). Optionally Nelder–Mead over $(\Omega_a, \Omega_b, \omega_b, T)$ as in the CZ workflow.
3. **Score:** logical block of $U_I(T) = e^{+i\hat{H}_0 T} U(T)$; average fidelity vs X maximized over the free virtual- Z frame; leakage $L_1 = 1 - \frac{1}{2} \text{Tr}(PU^\dagger PUP)$ reported separately.
4. **Axis check:** verify the gate phase follows ϕ_b ($\Phi = -\phi_b$) and is independent of ϕ_a .
5. **Population plots:** $P(0 \rightarrow k)$ and $P(1 \rightarrow k)$ vs t for the top- n participating levels (the multi-photon fingerprint: transient population of $\{4, 5\}$ returning to $\{0, 1\}$ at T).

7 Step 4 (recommended additions): robustness and open system

(i) **Offset-charge robustness:** repeat Step-3 scoring over $n_g \in [0, 1)$; the drift of the calibrated ω_b vs n_g sets the recalibration cadence in experiment. (ii) **Flux jitter:** fidelity vs small $\delta\varphi_{\text{ext}}, \delta\vartheta_{\text{ext}}$ excursions. (iii) **Parameter dispersion:** $\pm \text{few-}\% E_J$, and the fit uncertainties of the SI Note 11 set. (iv) **Open system:** repeat the final configuration with `mesolve` and per-level T_1, T_φ collapse operators; the coherence of $|4\rangle, |5\rangle$ is the dominant unknown and should be bracketed (e.g. 1–100 μs) until measured. (v) **Honest reporting:** every fidelity is quoted together with the idle-protection metrics $\mathcal{P}(\epsilon^*)$ — a fast gate on an unprotected qubit is not the objective.

8 Deliverables and milestones

- M0** Validated four-mode implementation (Step-0 checklist passed; convergence protocol documented).
- M1** Degeneracy-lifting maps $\delta_{01}, \delta_{23}, \delta_{45}$ and matrix-element maps over $(\epsilon_J, \epsilon_{LK}) \in [0, 0.15]^2$; suppression factor of the multimode model relative to the 1-mode caricature.
- M2** Sweet-spot report: acceptance window in $(\epsilon_J, \epsilon_{LK})$, re-centered bias, tone placement, and predicted Ω_X with protection budget.
- M3** Calibrated time-domain gate at the sweet spot: fidelity (up to virtual- Z), leakage, population plots, axis check.
- M4** Robustness/open-system appendix (Step 4).

Risks and notes

- **Truncation bias** is the main numerical risk (splittings are small); the Step-0 convergence protocol is not optional.
- **Model risk:** array modes near 5–6 GHz (SI Note 11) are not in the four-mode model; transitions landing near them should be flagged, not trusted.
- **Parameter risk:** the SI Note 11 fit is one device; rerun M1 at the Extended-Data-Fig.-3 parameter set to test conclusions’ stability.
- **Interpretation:** asymmetry cannot be flux-tuned away (unequal first-harmonic phasors leave a floor $E_{1J} \gtrsim 2\epsilon_J E_J$), which is what makes it a *reliable* resource; but for the same reason it cannot be switched off — the protection cost is permanent, hence the acceptance-window framing.